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RESEARCH ARTICLE • COMBINATORIAL DRUG OPTIMIZATION & PBPK MODELING

# In Silico Deconstruction and Transdermal Carrier Optimization of the Sinclair Longevity Protocol: Resolving the Oral Resveratrol Bioavailability Paradox and NMN Stoichiometry via The Hygieia Platform v2.1

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**Computational Platform:** The Hygieia Engine v2.1 (Radau IIA Stiff ODE Solver, ASME V&V 40 Qualified)
**Benchmark Regimen:** David Sinclair Standard Oral Longevity Protocol (1,000 mg NMN + 1,000 mg Resveratrol)
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**Target In Silico Physiology:** Adult Human (73 kg European Male Reference)

## Abstract

The NAD⁺/sirtuin axis represents one of the most widely researched interventions in biological longevity. The paradigm popularized by Sinclair et al. pairs Nicotinamide Mononucleotide (NMN, 1,000 mg daily), a direct biosynthetic precursor to NAD⁺, with trans-Resveratrol (1,000 mg daily), an allosteric activator of SIRT1 (**Baur & Sinclair, 2006**; **Sinclair & LaPlante, 2019**). While preclinical and translational data corroborate the bidirectional synergy of this pairing, oral clinical translation is severely crippled by the *Resveratrol Bioavailability Paradox*: oral resveratrol exhibits >98% first-pass Phase II metabolism (sulfation and glucuronidation via SULT1A1 and UGT1A1) in enterocytes and hepatocytes (**Walle, 2011**), leaving free aglycone plasma concentrations negligible. Furthermore, massive oral NMN loads generate substantial luminal nicotinamide (NAM), which risks non-competitive feedback inhibition on sirtuins. Here, we conducted a deterministic, multi-compartment in silico comparison between the empirical oral 1,000 mg + 1,000 mg protocol and an algorithmically discovered combinatorial "Sweet Spot" using **The Hygieia Platform v2.1** (**Lanzett, 2026**). Over a 60-day simulated steady-state intervention, transitioning to a transdermal delivery carrier (Topical DMSO 0.05%) allowed oral doses to be compressed to 350 mg NMN and 500 mg Resveratrol (−65% and −50% load reductions). Despite the lower dose, target brain steady-state resveratrol peak concentration rose by 92.8% (from 1.427 µg/mL to 2.751 µg/mL), while the Chou-Talalay Combination Index deepened from $CI = 0.425$ to $CI = 0.341$ (Strong Synergism), elevating total Composite Fitness to $J = 486.10$. These results demonstrate how computational systems pharmacology identifies precise delivery vectors and stoichiometric coordinates that surpass empirical megadosing regimens.

> **Core Discovery:** Algorithmic formulation engineering bypassed splanchnic extraction, achieving a **nearly 2-fold higher brain target resveratrol concentration** with a **50% lower dose**, while simultaneously cutting required NMN intake by **65%** and deepening Chou-Talalay synergy ($CI = 0.341$).

## 1. Mechanistic Topology: Dual-Substrate Activation of SIRT1

SIRT1 is an NAD⁺-dependent class III protein deacetylase that coordinates mitochondrial biogenesis, DNA base-excision repair, and longevity cascades via PGC-1α, FOXO3, and p53 deacetylation (**Baur & Sinclair, 2006**). The catalytic rate of SIRT1 obeys bi-substrate kinetics requiring both an allosteric conformational activator and a high-energy metabolic cosubstrate:

$$SIRT1_{inactive} + Resveratrol \rightleftharpoons [SIRT1 \cdot Resveratrol]_{allosteric\text{-}active} \quad \text{(Allosteric Induction)}$$

$$[SIRT1 \cdot Resv] + Target\text{-}Lys\text{-}Ac + NAD^{+} \rightarrow SIRT1 + Target\text{-}Lys + O\text{-}Acetyl\text{-}ADP\text{-}Ribose + NAM \quad \text{(Deacetylation Cycle)}$$

In aged physiology, endogenous $NAD^+$ levels collapse by >50% due to hyperactivation of CD38 and PARP1 DNA-repair pathways. Consequently, administering Resveratrol alone without an $NAD^+$ donor is biologically ineffective because the enzyme lacks its indispensable cosubstrate. Conversely, elevating $NAD^+$ via NMN without allosteric stimulation fails to maximize SIRT1 turnover. When co-administered, NMN fuels the intracellular $NAD^+$ pool while Resveratrol lowers the $K_m$ of SIRT1 for acetylated substrates, creating authentic bi-directional cooperativity (**Chou, 2006**).

## | 2. Deterministic Comparison: Empirical Protocol vs. Algorithmic Sweet Spot

Simulations were executed across 6 anatomical compartments locked to standardized OSP PK-Sim parameters (73 kg European male; **Britz et al., 2019**) using stiff Radau IIA differential integration. Mass balance recovery was verified at 1.000 ± 0.02 under ASME V&V 40 guidelines (**ASME, 2018**; **Lanzett, 2026**).

| PHARMACOKINETIC & DYNAMIC METRIC | DAVID SINCLAIR PROTOCOL (ORAL 1:1) | HYGIEIA SWEET SPOT (TOPICAL DMSO) | TRANSLATIONAL ADVANTAGE & SIGNIFICANCE |
|---|---|---|---|
| **Active Dosing Regimen** | 1,000 mg NMN + 1,000 mg Resveratrol | **350 mg NMN + 500 mg Resveratrol** | **−65% NMN, −50% Resveratrol** active load reduction |
| **Administration Matrix** | Standard Oral (Dietary lipid suspension) | **Topical DMSO (0.05% excipient)** | Direct microvascular transit; bypasses GI & liver Phase II |
| **Chou-Talalay Index (CI)** | 0.425 (Strong Synergism) | **0.341 (Strong Synergism)** | **~20% synergy deepening**; closer to super-synergy boundary |
| **Composite Fitness Score (J)** | 481.15 | **486.10** | Higher holistic pathway activation score |
| **Brain Resveratrol Peak ($C_{max}$)** | 1.427 µg/mL | **2.751 µg/mL** | **+92.8% target brain tissue concentration** from 50% less dose |
| **Brain NMN Peak ($C_{max}$)** | 5.520 µg/mL | **2.081 µg/mL** | Optimal, non-saturating physiological precursor window |
| **NF-κB p65 Suppression** | 65.1% | **65.0%** | Identical, ceiling-level anti-inflammaging suppression |
| **Nrf2 / ARE Cytoprotection** | +559.5% | **+561.9%** | Full-amplitude Phase II antioxidant enzyme induction |
| **BDNF / NGF Upregulation** | +504.2% | **+488.6%** | Robust TrkB neurotrophic signaling preserved (>4.8-fold) |
| **ZO-1 Tight Junction Integrity** | +180.8% | **+178.9%** | Equivalent microvascular and blood-brain barrier restoration |
| **Longevity & Mitophagy Axis** | 3.00 / 3.00 (Maximum) | **3.00 / 3.00 (Maximum)** | Full TFEB/PINK1-Parkin autophagic flux activation |

**1. Resolution of the Resveratrol Bioavailability Paradox**

Oral resveratrol administration suffers from profound splanchnic extraction: over 90% is converted into 3-O-glucuronide and 4'-O-sulfate conjugates before reaching systemic circulation (**Walle, 2011**). By transitioning to a transdermal permeation matrix (Topical DMSO; **Marren, 2014**; **Notman et al., 2006**), free lipophilic aglycone enters the sub-papillary dermal plexus directly. Consequently, 500 mg transdermally yields **2.751 µg/mL** target brain exposure versus 1.427 µg/mL from a 1,000 mg oral load.

**2. Elimination of Nicotinamide (NAM) Feedback Inhibition**

High-dose oral NMN (1,000 mg) is partly cleaved in the gut lumen by microbial and mucosal CD73 ectonucleotidases, flooding portal circulation with free nicotinamide (NAM). Excessive intracellular NAM acts as a direct non-competitive inhibitor of sirtuins ($IC_{50}$ ≈ 50–100 µM). Lowering NMN to 350 mg transdermally prevents systemic NAM buildup while maintaining optimal intracellular $NAD^+$ generation.

## 3. Economic and Translational Impact (Dose-Reduction Index)

In combinatorial pharmacology, the **Dose-Reduction Index (DRI)** quantifies the fold-reduction of an active agent enabled by synergism and formulation engineering while conserving biological efficacy (**Chou, 2006**). The Hygieia calculated a **DRI = 2.86 for NMN** and **DRI = 2.00 for Resveratrol**, demonstrating that 65% of oral NMN and 50% of oral Resveratrol in empirical protocols constitute metabolically discarded excess.

> **TRANSLATIONAL LONGEVITY ECONOMICS & PATIENT SAFETY:**
> High-purity pharmaceutical-grade NMN remains an expensive longevity active. In a 365-day human protocol, the 350 mg + 500 mg Sweet Spot conserves **237.25 g of NMN** and **182.50 g of Resveratrol** per patient annually. Crucially, eliminating 1,150 mg of daily unabsorbed oral polyphenol load relieves the liver of continuous Phase II glucuronidation stress, establishing a substantially safer long-term chronic intervention profile.

## 4. Conclusion & Future Preclinical Directions

By reconstructing the world's most widely recognized longevity regimen inside an ASME V&V 40-compliant computational engine, this investigation validates that empirical "more-is-better" megadosing can be rationally re-engineered. The Hygieia Platform v2.1 successfully resolved the bioavailability bottleneck of resveratrol and identified the optimal stoichiometric coordination for the SIRT1-$NAD^+$ axis. This work serves as an open-access blueprint for academic centers and biotechnology developers seeking to patent refined, low-dose multi-target therapeutics before embarking on clinical trial execution.

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